

Computer Practical Parcel Modelling

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1 Overview

In this computer practical, you will use a parcel model to investigate how water vapour condenses onto aerosol particles to form cloud droplets. As in previous practicals, the model advances a set of coupled ordinary differential equations in time.

By the end of the practical you should be able to:

- explain why a rising air parcel cools and eventually forms cloud;
- describe how vertical velocity and initial humidity affect cloud formation;
- explain how aerosol number, size-distribution width and particle size affect cloud droplet number;
- configure, run and analyse a compiled environmental model.

2 Model formulation

This model calculates the properties of an air parcel as it moves vertically through the atmosphere. Atmospheric pressure changes with height; expansion associated with decreasing pressure cools a rising parcel and can eventually lead to condensation. The model solves several coupled equations. A simplified summary is

$$e_0 = 610.7 \exp \left(\frac{L_v}{R_v} \left[\frac{1}{273.15} - \frac{1}{T} \right] \right), \quad (1)$$

$$e_v = e_0 \exp \left(\frac{2\sigma_w}{R_v T \rho_l r_i} \right), \quad (2)$$

$$\frac{dP}{dt} = -\rho_a g w, \quad (3)$$

$$\frac{dT}{dt} = \frac{R_a T}{c_p P} \frac{dP}{dt} + \frac{L_v}{c_p} \frac{dw_l}{dt}, \quad (4)$$

$$\frac{dm_i}{dt} = 4\pi r_i D_v \left(w_{v,\infty} \rho_a - \frac{e_v(T)}{R_v T} \right), \quad (5)$$

$$\frac{dw_l}{dt} = \sum_i^N N_i \frac{dm_i}{dt}, \quad (6)$$

$$\frac{dw_{v,\infty}}{dt} = -\frac{dw_l}{dt}. \quad (7)$$

Here e_0 and e_v are the equilibrium vapour pressures over a flat and curved surface respectively; L_v is the latent heat of vaporisation; R_v and R_a are the specific gas constants for water vapour and air; T is temperature; σ_w is the surface tension of water; ρ_l and ρ_a are the densities of liquid water and air; r_i is the radius of an aerosol particle or droplet; P is pressure; g is gravity; c_p is the specific heat capacity of air; m_i is the mass of a single aerosol particle or droplet; $w_{v,\infty}$ is the water-vapour mixing ratio; N_i is the number of aerosol particles or droplets; w is the vertical wind; and D_v is a diffusivity.

These are the equations that the Fortran model will solve for us.

3 Aerosol size distributions

Atmospheric aerosol particle sizes are often represented using log-normal distributions. Three parameters describe a log-normal mode: number concentration N , median diameter $\bar{D}$ and geometric standard deviation σ_g . Atmospheric aerosol populations commonly contain several modes, such as nucleation, accumulation and coarse modes, so the input file contains three modes for each aerosol type.

In the Marine Cloud Brightening geoengineering concept, adding sea-salt aerosol to the background aerosol may alter cloud droplet number and cloud reflectivity. This is the idea we will explore in the practical.

The parcel model is written in Fortran and follows an adiabatic parcel along a vertical coordinate. It generates output in NetCDF format, a common data format in atmospheric and environmental sciences. A Python script reads this output and produces the plots used in the exercises.

4 Log in, download and compile the code

Use the same two-window SSH/SFTP workflow as in *Tools of the Trade*. On campus, the teaching-server address is 130.88.66.57; otherwise use the server details supplied for the course. Replace YOUR_USERNAME with your own username.

In your SSH window:

```
ssh -Y YOUR_USERNAME@130.88.66.57
```

Open a second terminal window for file transfer:

```
sftp YOUR_USERNAME@130.88.66.57
```

Your password will not be displayed while you type it.

If you have not already downloaded the practical code, type in the SSH window:

```
git clone https://github.com/EnvModelling/bin-microphysics-model
```

This downloads the repository into your current working directory.

Move into the directory, inspect its contents and compile the Fortran code:

```
cd bin-microphysics-model
ls
make
```

You can type `cd bin-` followed by the Tab key to auto-complete the directory name. The `make` command compiles the Fortran source into machine code that the computer can run.

5 Viewing and editing the input file

The model input file is called `namelist.in`. Open it in nano with line numbers displayed:

```
nano -l namelist.in
```

Save and exit using Ctrl-X, then Y, followed by Enter.

Rather than relying only on line numbers, you can search within nano using Ctrl-W. For example, search for `winit`, `rhinit` or `n_aer1`. In the current version, `winit` is around line 18, `rhinit` around line 28, and the aerosol size-distribution parameters are around lines 161–168.

The arrays `n_aer1`, `d_aer1` and `sig_aer1` contain 3×2 values: three log-normal submodes for each of two externally mixed aerosol types. The first three values describe the background aerosol; the second three describe sea-salt aerosol.

6 Running, plotting and saving your results

After the code has been compiled, the normal workflow is:

1. edit `namelist.in` to configure the model;
2. run the model using `./run.sh`;
3. plot the NetCDF output using `python/example_plot_bmm2.py`;
4. download the resulting PNG in the SFTP window using a descriptive local filename.

If you change any Fortran source code, run `make` again before running the model. You do *not* need to recompile after changing only `namelist.in`.

The model writes its NetCDF output to your temporary directory, normally

```
/tmp/YOUR_USERNAME/output1.nc
```

The plotting script reads that file and writes

```
/tmp/YOUR_USERNAME/vertical.png
```

The updated plotting script also prints the name of the NetCDF file it reads and the PNG file it saves, so you can check that you are working with the expected files.

The remote file is always called `vertical.png`, so the next plot will overwrite it. Download each figure immediately and give the local copy a descriptive name.

In SFTP, the general form is

```
get REMOTE_FILE LOCAL_FILENAME
```

For example:

```
get /tmp/YOUR_USERNAME/vertical.png parcel_default.png
```

In SFTP, `$USER` is not expanded by the remote shell, so type your actual username in the remote path.

Table 1: Parameters used in this practical. The aerosol arrays list the three background modes followed by the three sea-salt modes.

Variable	Default value	Description
<code>winit</code>	0.3	parcel vertical velocity (m s^{-1})
<code>tinit</code>	290	initial temperature (K)
<code>pinit</code>	100000	initial pressure (Pa)
<code>rhinit</code>	0.65	initial relative humidity (fraction)
<code>n_aer1</code>	46.64e6, 153.42e6, 166.77e6 0e6, 0e6, 0.e6	number concentration in each mode (kg^{-1})
<code>d_aer1</code>	18e-9, 39e-9, 154.e-9 1e-6, 1e-6, 100.e-9	number-median dry diameter in each mode (m)
<code>sig_aer1</code>	0.348, 0.354, 0.465 0.3, 0.3, 0.5	$\ln \sigma_g$ for each mode (distribution width)

7 Experiments

Keep a practical record. Save each figure and make brief notes as you go. Record the parameter values used for each simulation and what changed in the output. You may be asked to interpret these experiments in the assessment.

7.1 Temperature lapse rate of the parcel

The first simulation is the default case: background aerosol, a parcel starting at the ground and rising at 0.3 m s^{-1} to a height of 1200 m. Run it with:

```
./run.sh
```

Check that the output file is present:

```
ls -lh /tmp/$USER/
```

Here \$USER is expanded automatically by the Linux shell to your username.

Plot the results:

```
python3 python/example_plot_bmm2.py
```

The script should print something similar to:

```
Reading model output: /tmp/YOUR_USERNAME/output1.nc  
Saved plot: /tmp/YOUR_USERNAME/vertical.png
```

In the SFTP window, download the figure with a descriptive local filename:

```
get /tmp/YOUR_USERNAME/vertical.png parcel_default.png
```

The figure shows the properties of the parcel along the vertical coordinate and will be discussed in the practical.

Exercise. From the top-right plot, estimate how rapidly temperature decreases with height up to about 800 m altitude. This lapse rate is characteristic of dry air rising approximately adiabatically.

Exercise. What happens to the rate of cooling above about 800 m? Why?

Make a note of the liquid-water mixing ratio at cloud top and the approximate height at which cloud droplets first form.

7.2 Effect of vertical wind speed

Vertical velocity controls how rapidly the parcel cools with time and how quickly supersaturation develops, which can strongly affect droplet activation. Change `winit` (around line 18 in the current `namelist.in`).

Run simulations at ten times less and ten times more than the default vertical velocity:

```
winit=0.03,
```

and

```
winit=3.0,
```

Open the `namelist` using:

```
nano -l namelist.in
```

After each change, run the model and plotting script:

```
./run.sh  
python3 python/example_plot_bmm2.py
```

Then download the plots in SFTP using different local filenames:

```
get /tmp/YOUR_USERNAME/vertical.png parcel_w0p03.png
```

and

```
get /tmp/YOUR_USERNAME/vertical.png parcel_w3p0.png
```

Exercise. How does the lapse rate between 0 and 800 m compare between these simulations?

Exercise. Look at both the liquid-water mixing ratio (top left) and CDNC (Cloud Drop Number Concentration). What effect does changing vertical velocity have? Why?

Exercise. Approximately what height does cloud start to form in these simulations? Why?

7.3 Increasing humidity (water vapour)

Return to the committed default state of the model:

```
git reset --hard
```

Find `rhinit` (around line 28 in the current `namelist.in`) and change it to

```
rhinit=0.95,
```

Run the model and generate the plot:

```
./run.sh
python3 python/example_plot_bmm2.py
```

Download it in SFTP as:

```
get /tmp/YOUR_USERNAME/vertical.png parcel_rh0p95.png
```

Exercise. How did increasing the initial humidity change the result? Why?

7.4 Adding sea-salt aerosol

From the main `bin-microphysics-model` directory, return to the default:

```
git reset --hard
```

We will now investigate three properties of an added sea-salt mode: number concentration, distribution width and particle size. The simulations are cumulative: **do not reset the model between runs 1–3.**

Table 2: Sea-salt experiments.

Run	Sea-salt number	Median diameter	$\ln \sigma_g$
Default	0	100 nm	0.5
1	$1000 \times 10^6 \text{ kg}^{-1}$	100 nm	0.5
2	$1000 \times 10^6 \text{ kg}^{-1}$	100 nm	0.01
3	$1000 \times 10^6 \text{ kg}^{-1}$	800 nm	0.01

1. **Add sea-salt aerosol.** Change the third sea-salt number concentration in `n_aer1` (around line 162) from zero to $1000 \times 10^6 \text{ kg}^{-1}$:

```
0e6, 0e6, 1000.e6,
```

Keep the default third-mode diameter (`100.e-9`) and width (`0.5`). Run and plot:

```
./run.sh
python3 python/example_plot_bmm2.py
```

Then download:

```
get /tmp/YOUR_USERNAME/vertical.png parcel_seasalt_number.png
```

2. **Narrow the sea-salt distribution.** Keep the sea-salt number concentration at $1000.e6$ and the diameter at $100.e-9$. Change only the third sea-salt value of `sig_aer1` (around line 168) from 0.5 to 0.01:

```
0.3, 0.3, 0.01,
```

Run and plot again, then download:

```
get /tmp/YOUR_USERNAME/vertical.png parcel_seasalt_narrow.png
```

3. **Increase the sea-salt particle size.** Keep the sea-salt number concentration at $1000.e6$ and `sig_aer1` at 0.01. Change only the third sea-salt value of `d_aer1` (around line 165) from $100.e-9$ to $800.e-9$:

```
1e-6, 1e-6, 800.e-9,
```

Run and plot again, then download:

```
get /tmp/YOUR_USERNAME/vertical.png parcel_seasalt_800nm.png
```

Exercise. Compare run 1 with the default simulation in Section 7.1. What is the effect of adding $1000 \times 10^6 \text{ kg}^{-1}$ sea-salt particles?

Exercise. Compare run 2 with run 1. What is the effect of narrowing the sea-salt size distribution while keeping its number concentration and median diameter fixed?

Exercise. Compare run 3 with run 2. What is the effect of increasing the sea-salt median diameter from 100 nm to 800 nm while keeping number concentration and distribution width fixed?

7.5 Bonus: Ice formation

From the main `bin-microphysics-model` directory, return to the default:

```
git reset --hard
```

Find `z_ctop` (around line 98 in the current `namelist.in`) and change the cloud-top height to 12000 m:

```
z_ctop=12000,
```

Run the simulation and make the plot:

```
./run.sh
python3 python/example_plot_bmm2.py
```

Download it as:

```
get /tmp/YOUR_USERNAME/vertical.png parcel_ice_12km.png
```

We will discuss the output in the practical.